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CHAPTER-1

COMPUTER GRAPHICS:

Computer graphics are graphics created using computers and more generally, the representation and manipulation of image data by a computer with help from specialized software and hardware.

The development of computer graphics has made computers easier to use and interact with. It also helps people understand and analyze different types of data more effectively. Computer graphics have greatly influenced many forms of media and have transformed animation, movies, and the video game industry.

APPLICATION OF COMPUTER GRAPHICS:

Computer graphics make computers easier to use and understand. They help people view and analyze information more clearly. Computer graphics are widely used in media and have changed the way animations, movies, and video games are created.

I. COMPUTER AIDED DESIGN (CAD):

A major use of computer graphics is in design, which is commonly called Computer- Aided Design (CAD). CAD is widely used to design buildings, cars, airplanes, ships, spacecraft, computers, textiles, and many other products.

In many CAD applications, objects are first shown as simple wireframe models to display their overall shape. CAD software usually provides multiple windows so designers can view and edit designs from different angles.

Animation is also used in CAD systems. Realistic 3D displays allow architects and clients to see how a building will look before it is built. Using virtual reality, designers can even take a virtual walk through rooms or around buildings to better understand the final design.

CAD software also helps architects experiment with interior layouts, furniture placement, and lighting in a three-dimensional environment.

II. PRESENTATION GRAPHICS:

Presentation graphics are used to create charts, graphs, and illustrations for reports and presentations. They help people understand information more easily by displaying data in a visual format. These graphics are commonly used to present financial, statistical, mathematical, scientific, and economic data in research reports, business reports, and information bulletins. Examples of presentation graphics include bar charts, line graphs, pie charts, and surface graphs. Three-dimensional (3D) graphs are also used to make presentations more attractive and to show data relationships in a clearer and more impressive way.

III. COMPUTER ART:

Computer graphics are widely used in both fine art and commercial art. Artists use various computer tools and software, such as paint programs, CAD software, desktop publishing applications, animation packages, and special-purpose hardware to create digital artwork.

Paint programs allow artists to draw and paint pictures directly on a computer screen using a graphics tablet and stylus, which can simulate different brush styles, sizes, and colors. Computer graphics are also used to create electronic art based on mathematical formulas and patterns. In some cases, visual effects are designed to match changes in music, combining visual and audio elements to create an artistic multimedia experience.

IV. ENTERTAINMENT:

Computer graphics are widely used in the entertainment industry to create movies, television shows, and music videos. They help produce realistic visual effects, animations, and special scenes that would be difficult or expensive to create in real life. Many films and TV series use computer graphics to enhance their visual quality. For example, computer-generated scenes were used in Star Trek: The Wrath of Khan. Computer graphics are also commonly used in music videos and for creating special effects such as morphing, where one person or object is smoothly transformed into another.

V. EDUCATION AND TRAINING:

Computer-generated models of physical, financial, and economic systems are often used as educational tools. These models help students and trainees better understand how different systems work. Visual representations, such as color-coded diagrams and simulations, make complex concepts easier to learn and understand. Computer graphics are also used in specialized training systems, such as simulators for ship captains, pilots, heavy equipment operators, and air traffic controllers. These simulators provide a safe and realistic environment for practice and skill development.

VI. VISUALIZATION:

Scientific and business visualization is an important application of computer graphics. It involves creating graphical representations of scientific, engineering, medical, financial, and business data to make the information easier to understand and analyze. Scientific visualization is used for scientific and medical data, while business visualization is used for commercial and industrial data. Different types of data, such as numerical values, vectors, and complex data structures, require different visualization methods. Common visualization techniques include color coding, contour plots, graphs, charts, surface rendering, and volume visualization. These methods help users identify patterns, trends, and relationships within large datasets more effectively.

VII. IMAGE PROCESSING:

Image processing is an application of computer graphics that involves modifying or analyzing existing images, such as photographs and television scans. The two main purposes of image processing are improving image quality and enabling computers to understand visual information. To perform image processing, a picture is first converted into a digital image file. Various techniques can then be applied to adjust colors, enhance shading, remove noise, and improve overall image quality. Image processing is often combined with computer graphics in many fields, including medicine. One important application is computer-aided surgery, where digital images help doctors plan and perform surgical procedures more accurately.

VIII. GRAPHICAL USER INTERFACE(GUI):

A Graphical User Interface (GUI) is a visual way of interacting with a computer using graphical elements such as windows, icons, buttons, and menus. Modern software applications commonly use GUIs because they are easier to use than text-based interfaces. A major part of a GUI is the window manager, which allows users to open and work with multiple windows at the same time. Each window can display different programs or information, including both graphical and text-based content. An icon is a small graphical symbol that represents a program, file, or function. Icons require less screen space than text descriptions and help users quickly recognize and access different options when they are well designed.